

Course: Advanced Topics in Artificial Intelligence and Machine Learning

Code: PROG74040

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Assignment/Lab 1: Spring 2025

Hands-On with Real Data: Autoencoders and Generative Adversarial Network

Objectives

- Implement and train Vanilla Autoencoder, Variational Autoencoder (VAE), and Generative Adversarial Network (GAN) models.
- Preprocess and explore MNIST, Fashion-MNIST, and a combined dataset (MNIST + Fashion-MNIST).
- Generate synthetic images using each trained model.
- Compare the quality and characteristics of generated images across datasets and models.
- Analyze the impact of training on single vs. combined datasets.
- Use visualization techniques to interpret results.

Dataset Description

You will work with two commonly used image datasets:

- **Fashion-MNIST:** Fashion item images (10 classes), 28×28 pixels, 60,000 training and 10,000 test.
- **MNIST:** Handwritten digits (0–9), 28×28 pixels, 60,000 training and 10,000 test images
- **combined dataset:** You will also create a **combined dataset** by concatenating the MNIST and Fashion-MNIST sets.

Instructions

Step 1: Data Loading and Preprocessing

- Load MNIST and Fashion-MNIST datasets using your preferred deep learning framework (TensorFlow/Keras or PyTorch).
- Normalize the images.
- Create a combined dataset by concatenating the MNIST and Fashion-MNIST training sets.
- Explore the datasets using visualization techniques (e.g., sample images, class distributions).
- Comment on differences between the datasets and potential challenges when combining them.

Step 2: Train Vanilla Autoencoder Models

(1 pt.)

- Implement a Vanilla Autoencoder suitable for 28×28 grayscale images. Use a 2D latent space.
- Train it on MNIST, Fashion-MNIST, and the combined dataset MNIST
- Visualize the latent space representation of the test data using scatter plots colored by class.
- Generate a grid of synthetic images after training completion for each dataset.
- Save the trained models.

Step 3: Train Variational Autoencoder (VAE) Models

(2 pts.)

- Implement a Variational Autoencoder (VAE) suitable for 28×28 grayscale images.
- Train it on MNIST, Fashion-MNIST, and the combined dataset.
- Visualize the latent space representation of the test data using scatter plots colored by class.
- Generate a grid of synthetic images after training completion for each dataset.
- Save the trained models.

Step 4 Train Generative Adversarial Networks (GANs)

(2 pts.)

- Implement a GAN architecture suitable for 28×28 grayscale images.
- Train the GAN on MNIST, Fashion-MNIST, and the combined dataset.
- At regular intervals (e.g., every n epochs), save generated images to observe training progress.
- Generate a grid of synthetic images after training completion for each dataset.
- Save the trained models.

Step 5 Comparison and Evaluation

(2 pts.)

- Compare the synthetic images generated by the Vanilla Autoencoder, VAE, and GAN models on each dataset.
- Discuss how training on the combined dataset affected the quality and diversity of generated images.
- Provide visual examples supporting your comparison
- Which model produced the best quality synthetic images? Explain why you think so.
- What challenges did you face when training GANs versus Autoencoders?
- How useful is the latent space visualization in understanding VAEs?

The overall organization and Clarity of your solution

(2 pts.)

- Ensure your solution is well-organized, clearly commented, and easy to follow
- In this lab the only limit is your imagination. Be creative \o/ (+2 pts.)

What to hand in?

Read the Assignment/Lab instructions section on the course shell.